

Lab 2: Configuring Security Groups and Network ACLs

Duration: 30 minutes · **Region:** us-east-2 (Ohio)

Objectives

- Build a VPC with two public subnets, an Internet Gateway and a route table
- Create a web security group and a database security group
- Add a rule to a network ACL and compare it with a security group

What you build (used by Labs 3–7):

```
VPC userN-vpc 10.0.0.0/16
├─ Public subnet 1 (us-east-2a) — web server (Lab 3)          security group: userN-web-
sg (80, 22 from anywhere)
├─ Public subnet 2 (us-east-2b) — web server 2 (Lab 4), RDS (Lab 7)  userN-db-sg (3306
from userN-web-sg)
└─ Route table: 0.0.0.0/0 → Internet Gateway
```

Exercise 2.1: Create the VPC (8 min)

1. Search **VPC** → **Create VPC** → choose **VPC and more**.
2. Settings:

Setting	Value
Name tag auto-generation	userN
IPv4 CIDR block	10.0.0.0/16
Number of Availability Zones	2
Number of public subnets	2
Number of private subnets	0
NAT gateways	None
VPC endpoints	None

3. **Create VPC**. Wait for the green checks, then **View VPC**.

Exercise 2.2: Enable Public IPs and Check the Route (5 min)

1. **Subnets** → for **each** of your two userN-subnet-public... subnets: select it → **Actions** → **Edit subnet settings** → check **Enable auto-assign public IPv4 address** → **Save**. (The wizard does not

do this for you.)

2. **Route tables** → open `userN-rtb-public` → **Routes** tab. You will see `10.0.0.0/16 → local` and `0.0.0.0/0 → igw-...`. That second route is what makes the subnets public.

Exercise 2.3: Web Security Group (5 min)

1. **Security groups** → **Create security group**.
2. Name `userN-web-sg`, description `web server`, **VPC:** `userN-vpc` (not the default).
3. **Inbound rules** → add two:

Type	Source
HTTP	Anywhere-IPv4
SSH	Anywhere-IPv4

4. **Create security group**.

Exercise 2.4: Database Security Group (4 min)

1. **Create security group** again: name `userN-db-sg`, description `database`, VPC `userN-vpc`.
2. **Inbound rules** → add one: Type **MYSQL/Aurora**, Source: start typing `userN-web-sg` and select it.
3. **Create security group**.

The database trusts *whatever server carries the web security group*, not an IP address. That is the standard AWS pattern.

Exercise 2.5: Network ACL (8 min)

1. **Network ACLs** → open the one whose VPC is `userN-vpc`. Look at **Inbound rules**: rule 100 allows all traffic; the last `*` rule denies everything else. This is the default NACL — wide open; the security groups do the real filtering.
2. **Edit inbound rules** → **Add new rule**:

Rule number	Type	Source	Allow/Deny
50	SSH (22)	<code>198.51.100.0/24</code>	Deny

3. **Save changes**. Rules are evaluated lowest number first, so rule 50 blocks SSH from that range before rule 100 allows everything.

A security group can only *allow*; a network ACL can also *deny*, and it applies to every instance in the subnet.

Checkpoint

- `userN-vpc` with 2 subnets, **auto-assign public IP enabled on both**
- `userN-web-sg` (80, 22) and `userN-db-sg` (3306 from web-sg)
- Default NACL has your Deny rule 50

Knowledge Check

1. Which route makes a subnet public?
2. Stateful or stateless: security group? network ACL?